

Introduction to Quantitative Biology

Lecture Notes: Positive feedback and bistability

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1. Biological question: can the same circuit produce different cell states?

In the previous lectures, we reduced gene regulation to a production-removal equation and asked how fast a system approaches a steady state. In that simple setting, one fixed environment gave one stable steady state.

Here we ask a different question:

Can genetically identical cells, experiencing the same external environment and carrying the same gene regulatory network, stably occupy different phenotypic states?

The answer can be yes. A gene regulatory network with sufficiently strong **positive feedback** can support more than one stable fixed point. The state reached by a cell can then depend on where the cell started, or on the sequence of environments it experienced previously.

This gives three linked ideas:

positive feedback $\implies$ bistability $\implies$ history dependence / memory.

The lac utilization network in *E. coli* provides a concrete biological example. To make the biological components correspond directly to the variables used below, we write the feedback loop as

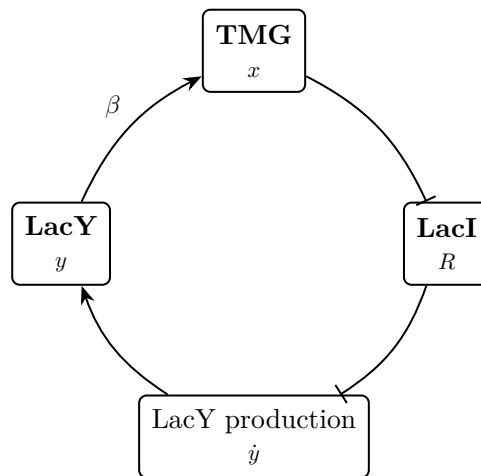


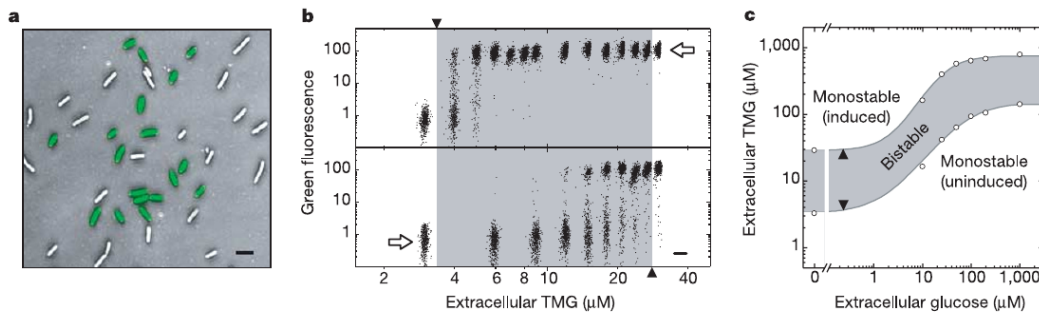
Figure 1. The lac positive-feedback loop. A blunt-ended connector denotes repression; an arrow denotes activation. The symbols x and y are the variables used in the model below. Thus LacY increases TMG uptake, intracellular TMG inactivates LacI, reduced LacI repression increases lacY production, and the loop closes as more LacY is produced. The parameter β will quantify how strongly LacY feeds back onto intracellular TMG.

1.1 The experimental phenomenon: the present environment is not always enough

In the single-cell experiments of Ozbudak et al., cells were prepared with different lac-expression histories and then placed into media with defined TMG concentrations. At zero glucose, low TMG ($T < 3 \mu\text{M}$) leaves the system uninduced and high TMG ($T > 30 \mu\text{M}$) leaves it induced. Between these values, however, the same extracellular TMG concentration can support two different expression states:

$$3 \mu\text{M} < T < 30 \mu\text{M}.$$

Cells that enter this interval from the uninduced state tend to remain OFF, whereas cells that enter it from the induced state tend to remain ON. This is the phenomenological observation that the model must explain.



Paper Figure 2. Experimental hysteresis and bistability in the lac system. The experiments compare cells with different initial lac-expression states and map the switching thresholds as extracellular TMG and glucose are varied. Adapted from Ozbudak et al. (2004), Fig. 2.

The central goal of this lecture is to understand this behavior **mathematically**. We will use **graphical analysis**, through production-removal rate plots and bifurcation diagrams, to analyze the model.

2. Start from the familiar production-removal equation

Let y denote the concentration or normalized expression level of the protein of interest. A generic one-dimensional gene-expression model has the form

$$\frac{dy}{dt} = \text{production rate} - \text{removal rate}.$$

Write the removal rate as $k_y y$:

$$\frac{dy}{dt} = p(\cdot) - k_y y, \tag{1}$$

where $p(\cdot)$ contains the regulatory dependence of production on the state of the system.

Define the characteristic removal time

$$\tau_y \equiv \frac{1}{k_y}.$$

Multiplying Eq. (1) by τ_y gives

$$\tau_y \frac{dy}{dt} = \tau_y p(\cdot) - y.$$

Now define a rescaled production function

$$P(\cdot) \equiv \tau_y p(\cdot).$$

Then

$$\boxed{\tau_y \dot{y} = P(\cdot) - y} \tag{2}$$

This rewriting is useful because the removal term is now simply the straight line

$$R(y) = y.$$

The production curve and the removal line therefore have the same vertical units. Their intersections will directly give the fixed points.

Notation warning. In this lecture we follow the notation of the lac model. Here α will denote a maximal lac-expression level, and β will denote the strength of TMG transport/feedback. These symbols are not being used in exactly the same way as in the previous lecture's generic production-removal notation.

3. Constructing the lac positive-feedback model

We keep two variables that describe the feedback loop:

- x : intracellular TMG concentration measured in units of the TMG-LacI dissociation constant x_0 ;
- y : LacY expression level, measured experimentally by green fluorescence.

The model is built by coarse-graining three biological processes. **The derivation from binding reactions to Eq. (6) is given in Appendix A. If the forms of Eqs. (3) and (4) are unfamiliar, it is recommended to read Appendix A before continuing.**

3.1 Intracellular TMG inactivates LacI

Let R_T be the total LacI concentration and R the active LacI concentration. Let x_0 denote the dissociation constant for TMG-LacI binding. Because x is measured in units of x_0 , a Hill form for the fraction of LacI that remains active is

$$\frac{R}{R_T} = \frac{1}{1 + x^n}. \quad (3)$$

Here n is the Hill coefficient. For the lac system, $n \approx 2$ is a useful approximation.

As x increases, active LacI decreases.

3.2 Active LacI represses LacY production

The normalized LacY production term is taken to be a decreasing hyperbolic function of active LacI:

$$P(R) = \frac{\alpha}{1 + R/R_0}, \quad (4)$$

where R_0 is the dissociation constant for LacI binding to the operator/promoter, and α is the maximal lac-expression level when repression is relieved.

Define the dimensionless repression factor

$$\rho \equiv 1 + \frac{R_T}{R_0}. \quad (5)$$

Using Eq. (3),

$$\frac{R}{R_0} = \frac{R_T/R_0}{1 + x^n} = \frac{\rho - 1}{1 + x^n}.$$

Therefore

$$1 + \frac{R}{R_0} = 1 + \frac{\rho - 1}{1 + x^n} = \frac{\rho + x^n}{1 + x^n},$$

and Eq. (4) becomes

$$\boxed{P(x) = \alpha \frac{1 + x^n}{\rho + x^n}} \quad (6)$$

so the LacY equation is

$$\boxed{\tau_y \dot{y} = \alpha \frac{1 + x^n}{\rho + x^n} - y.} \quad (7)$$

This production function has two useful limits:

$$P(0) = \frac{\alpha}{\rho}, \quad P(x \rightarrow \infty) = \alpha.$$

Thus ρ measures the fold repression between maximal and basal expression:

$$\rho = \frac{\text{maximal lac expression}}{\text{basal lac expression}}.$$

3.3 LacY imports TMG

TMG transport is described by

$$\boxed{\tau_x \dot{x} = \beta y - x.} \quad (8)$$

The term βy describes LacY-dependent TMG import, whereas $-x$ represents loss of intracellular TMG from the coarse-grained intracellular pool. TMG is not degraded by the cell, so its concentration falls mainly through dilution as the cell grows and divides. The parameter β therefore couples LacY expression back to intracellular inducer concentration.

Equations (7) and (8) form the two-variable positive-feedback model represented by the biological loop above.

4. Timescale separation reduces the model to one dimension

TMG transport and equilibration occur on the **millisecond scale**, whereas appreciable changes in LacY expression occur on the **minute scale**. We therefore assume

$$\tau_x \ll \tau_y.$$

On the timescale over which y changes appreciably, x has already relaxed close to its quasi-steady state. Set

$$\tau_x \dot{x} \approx 0,$$

so Eq. (8) gives

$$\boxed{x \approx \beta y.} \quad (9)$$

Substitute Eq. (9) into Eq. (7):

$$\boxed{\tau_y \dot{y} = \alpha \frac{1 + (\beta y)^n}{\rho + (\beta y)^n} - y.} \quad (10)$$

This is the one-dimensional dynamical system that we will analyze for the rest of the lecture.

Important: the Hill exponent acts on the product βy :

$$1 + (\beta y)^n,$$

not on $1 + \beta y$. The nonlinearity comes from the cooperative TMG-LacI response before the substitution $x = \beta y$.

What each parameter does

The reduced model separates conceptually different roles. τ_y sets the time scale but does not move the fixed points. α sets the maximal expression scale. ρ controls the strength of LacI repression and the basal expression α/ρ . n controls the steepness of the regulatory transition. Finally, β controls how strongly a given amount of LacY increases intracellular TMG; increasing β therefore strengthens the positive-feedback loop.

5. Graphical solution: production versus removal

Define

$$P_\beta(y) = \alpha \frac{1 + (\beta y)^n}{\rho + (\beta y)^n},$$

and

$$R(y) = y.$$

Then Eq. (10) becomes

$$\tau_y \dot{y} = P_\beta(y) - R(y). \quad (12)$$

Define the net rate

$$F(y; \beta) \equiv P_\beta(y) - y. \quad (13)$$

A fixed point y^* satisfies

$$F(y^*; \beta) = 0,$$

or equivalently

$$\boxed{P_\beta(y^*) = y^*}. \quad (14)$$

Thus every intersection of the sigmoidal production curve with the removal line is a fixed point.

The direction of motion can be read directly from the vertical distance between the curves:

$$P_\beta(y) > y \quad \Rightarrow \quad \dot{y} > 0,$$

whereas

$$P_\beta(y) < y \quad \Rightarrow \quad \dot{y} < 0.$$

So we do not need an explicit closed-form solution for $y(t)$ to understand the long-term behavior.

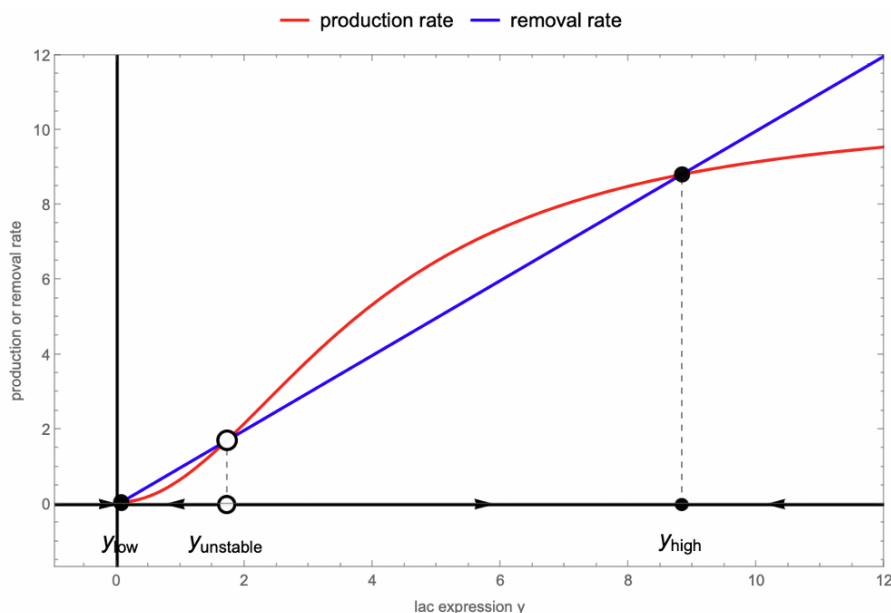


Figure 2. Reading dynamics directly from the rate plot. The intersections of production and removal are fixed points. The arrows drawn along the horizontal y -axis indicate the direction in which the system evolves over long times. Filled circles denote stable fixed points and the open circle denotes the unstable fixed point.

6. Bistability, basins of attraction, and memory

Figure 2 already contains the essential basin structure. The unstable fixed point y_u divides the horizontal axis into two basins of attraction:

$$\underbrace{y < y_u}_{\text{basin of attraction of } y_{\text{low}}} \quad \bigg| \quad \underbrace{y = y_u}_{\text{unstable separatrix}} \quad \bigg| \quad \underbrace{y > y_u}_{\text{basin of attraction of } y_{\text{high}}} .$$

For $y < y_u$, the arrows on the axis point toward y_{low} ; for $y > y_u$, they point toward y_{high} . At y_u , trajectories on either side move away from the fixed point. Therefore the outer intersections are stable and the middle intersection is unstable.

The same current parameter values can therefore support two different long-term states. Which state is reached depends on which side of y_u the system starts on. This is the first dynamical meaning of **memory**: the present state is not determined by the present environment alone, but also by the system's history.

The memory is not stored in the lifetime of any one LacY or LacI molecule. Individual molecules can turn over while the **feedback circuit continually regenerates the attractor**.

7. A control parameter moves the rate plot

To see how switching occurs, we do not draw only one rate plot. We draw a **family of rate plots** while varying one parameter.

For the lac circuit, β is a useful control parameter because extracellular TMG and glucose influence the effective TMG transport/feedback strength. At fixed α , ρ , and n , increasing β means that the same LacY level produces a larger intracellular TMG concentration:

$$x = \beta y.$$

Therefore the transition in $P_\beta(y)$ occurs at a lower value of y . Graphically, increasing β shifts the sigmoidal transition to the left while leaving

$$P_\beta(0) = \frac{\alpha}{\rho}$$

and

$$P_\beta(\infty) = \alpha$$

unchanged.

The system contains **two saddle-node bifurcations**. Since the bistable state already contains three fixed points, the clearest way to see the two folds is to start inside the bistable region and then vary β in either direction.

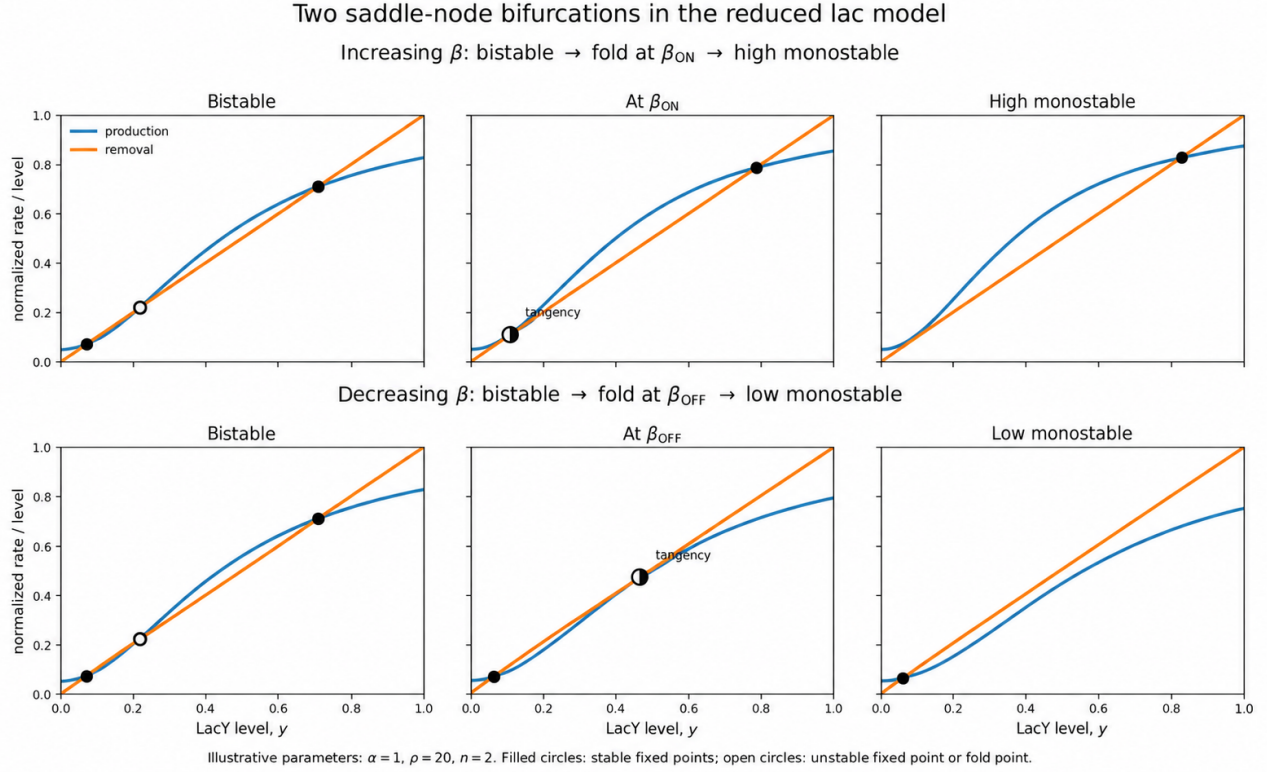


Figure 3. The two saddle-node bifurcations in the reduced lac model. Top row: starting from the three-fixed-point bistable state and increasing β , the low stable fixed point and the unstable fixed point meet at a tangency at β_{ON} ; for larger β , only the high stable fixed point remains. Bottom row: starting from the same bistable state and decreasing β , the high stable fixed point and the unstable fixed point meet at a tangency at β_{OFF} ; for smaller β , only the low stable fixed point remains. At each fold, one ordinary fixed point remains while the other two fixed points have merged at the tangency.

The key point is that moving away from the bistable region in either direction removes one stable-unstable pair at a saddle-node bifurcation, leaving a single stable state.

8. Saddle-node bifurcations: where two fixed points meet

A **bifurcation** is a qualitative change in the dynamical structure of a system as a control parameter is varied.

In the lac model, the relevant local event is a **saddle-node bifurcation**, also called a **fold bifurcation**. Figure 3 follows the system from the bistable region outward: a stable and an unstable fixed point meet at a tangency and disappear together.

At an ordinary fixed point,

$$F(y^*; \beta) = 0.$$

At the fold, two roots have merged into a double root, so we also require

$$\frac{\partial F}{\partial y}(y^*; \beta) = 0.$$

Therefore the bifurcation point satisfies

$$\boxed{F(y^*; \beta) = 0, \quad \frac{\partial F}{\partial y}(y^*; \beta) = 0.} \quad (15)$$

In the production-removal picture, these are exactly the **intersection and tangency conditions**:

$$\boxed{P_\beta(y^*) = y^*, \quad P'_\beta(y^*) = 1.} \quad (16)$$

Thus the graphical statement “the two curves just touch” corresponds to the intersection and tangency conditions at the switching threshold.

9. From a family of rate plots to a bifurcation diagram

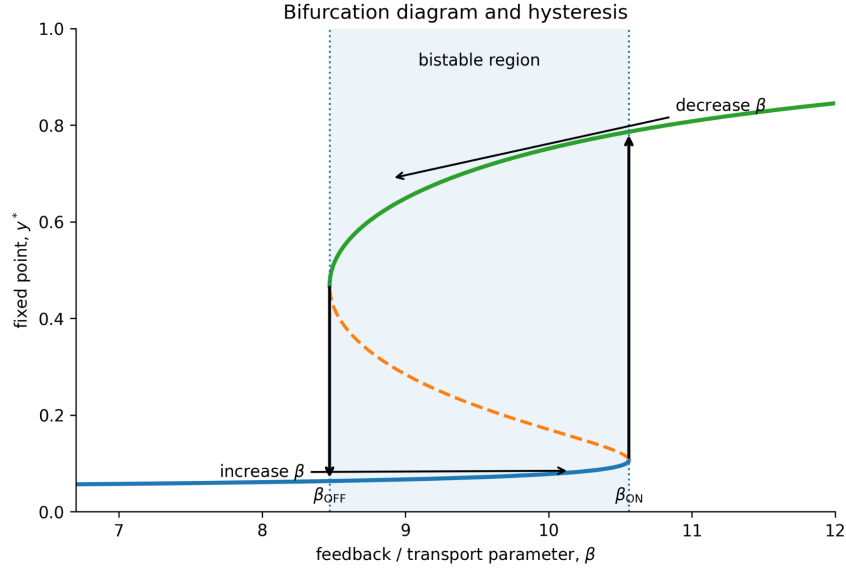
Figure 3 showed the two saddle-node bifurcations locally. We now follow the stable branches as the control parameter is changed in either direction.

Instead of redrawing the full production-removal curves for every value of β , we can record only their fixed points.

For each value of β :

1. find every intersection $P_\beta(y) = y$;
2. classify each intersection as stable or unstable;
3. plot the fixed-point value y^* against β .

The result is a **bifurcation diagram**.



Solid branches: stable fixed points; dashed branch: unstable fixed points. Threshold values are illustrative, not experimental fits.

Figure 4. Bifurcation diagram and hysteresis. The two folds delimit the interval in which both a low and a high stable state exist. The middle branch is unstable. The two fold points have different biological meanings.

9.1 Increasing the inducer: the ON threshold

Start on the low-expression branch and slowly increase β . The cell remains on the low stable branch even after a high stable branch has appeared. It does not switch simply because another stable state exists.

Eventually, at the upper threshold β_{ON} , the low stable fixed point collides with the unstable fixed point:

$$y_{\text{low}} + y_u \longrightarrow \text{no low fixed point.}$$

The OFF state has disappeared, so the trajectory must move to the high-expression branch.

9.2 Decreasing the inducer: the OFF threshold

Now start on the high-expression branch and slowly decrease β . The cell remains ON throughout part of the same parameter range.

Eventually, at the lower threshold β_{OFF} , the high stable fixed point collides with the unstable fixed point:

$$y_{\text{high}} + y_u \longrightarrow \text{no high fixed point.}$$

The ON state has disappeared, so the system jumps to the low-expression branch.

Because

$$\boxed{\beta_{\text{OFF}} < \beta_{\text{ON}},}$$

the switching point depends on the direction from which the parameter is approached. This is **hysteresis**.

Within

$$\beta_{\text{OFF}} < \beta < \beta_{\text{ON}},$$

the current environment alone is not enough to determine the state. The system also carries information about its history.

10. Why bistability is a form of cellular memory

The distinction between “two possible states” and “memory” is important.

Bistability means that two stable attractors coexist under the same parameter values. Hysteresis shows how a cell can enter one attractor and remain there as the environment is changed back. Together they generate a practical form of history dependence.

Suppose two cells are currently exposed to the same TMG concentration inside the bistable interval. One cell approached this environment from low TMG and can remain OFF. Another approached it from high TMG and can remain ON. Their genomes and present environment can therefore be the same, while their expression states differ because their trajectories through state space were different.

This is the dynamical answer to the opening question:

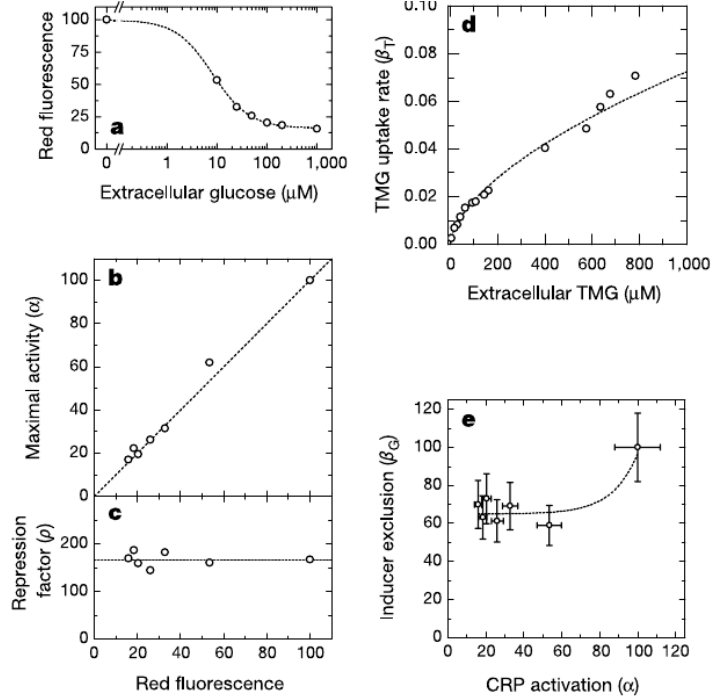
$$\boxed{\text{same genes} + \text{same present environment} \not\Rightarrow \text{same cell state}}$$

when the regulatory network contains multiple stable attractors.

11. From experiment to fitted parameters and a model test

Paper Figure 2 provides the experimental switching thresholds and fluorescence levels. These measurements were then used to connect the model parameters to the wild-type lac network.

11.1 Parameter fitting from the switching data



Paper Figure 3. Single-cell in vivo measurement of network parameters. Adapted from Ozbudak et al. (2004), Fig. 3.

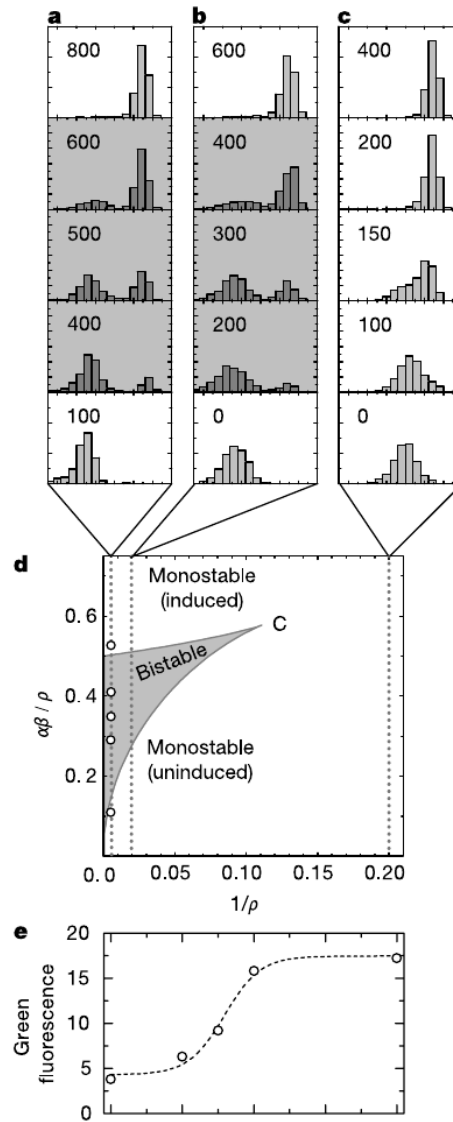
The model parameters were extracted by fitting to the measured fluorescence values at the switching thresholds. At each glucose condition, the ON/OFF switching thresholds together with the corresponding low- and high-expression levels constrain the fit. The maximal activity α follows CRP activity, whereas the repression factor is approximately

$$\rho \approx 170 \pm 30$$

across the glucose and TMG conditions measured in the wild-type network. The transport parameter can be separated into a TMG-dependent uptake term β_T and a glucose-dependent active-permease fraction β_G . Thus, in the wild-type parameter measurements, glucose changes α and β_G rather than ρ .

11.2 Testing a prediction by changing repression

The fitted model predicts that reducing the repression factor ρ shrinks the bistable region and can eventually replace the hysteretic response with a graded response.



Paper Figure 4. Model prediction followed by an experimental perturbation. Adapted from Ozbudak et al. (2004), Fig. 4.

To test this prediction, additional LacI-binding sites were introduced on multicopy plasmids, titrating LacI and lowering the effective repression factor from the wild-type value of about 170 to about 50 and 5. The $\rho \approx 50$ strain remained hysteretic, whereas the $\rho \approx 5$ strain showed the predicted graded response. In the model, the bistable region disappears near the critical repression factor $\rho \approx 9$.

This closes the experimental logic used in the lecture: measure the bistable response, fit the model parameters, use the model to predict a new perturbation, and then test that prediction experimentally.

12. Summary

The lecture begins from the same production-removal structure used previously,

$$\tau_y \dot{y} = P - y,$$

but positive feedback makes the production term depend on the system's own state. In the lac circuit, TMG inactivates LacI and active LacI represses LacY production. Combining these two binding relationships gives

$$P(x) = \alpha \frac{1 + x^n}{\rho + x^n}.$$

Because TMG transport equilibrates much faster than LacY expression, x can be replaced by its quasi-steady value $x \approx \beta y$. The two-variable feedback loop therefore reduces to

$$\tau_y \dot{y} = \alpha \frac{1 + (\beta y)^n}{\rho + (\beta y)^n} - y.$$

The rest of the analysis can be done graphically. Intersections of production and removal are fixed points, and the arrows on the y -axis show the direction of long-term motion. When the sigmoidal production curve crosses the removal line three times, the two outer intersections are stable and the middle one is unstable. That unstable point separates two basins of attraction, so two cells in the same present environment can approach different stable states depending on their histories.

Changing the control parameter β moves the production curve. Stable and unstable fixed points can collide and disappear through **saddle-node bifurcations**. Collecting the fixed points for different β values gives the bifurcation diagram, whose two fold points define distinct ON and OFF thresholds. The resulting hysteresis is the observable signature of history dependence.

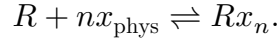
The experimental analysis then connects the model back to data. The switching measurements are used to infer the model parameters, with ρ approximately constant across glucose and TMG conditions in the wild-type network. A separate perturbation experiment lowers the effective LacI repression and tests the model prediction that sufficiently weak repression removes bistability and produces a graded response.

Appendix A. From binding reactions to the lac production function

This appendix derives Eqs. (3)-(6) from simple binding equilibria. The goal is to show where the Hill form for TMG-LacI regulation and the hyperbolic repression function for LacI-promoter binding come from.

A.1 TMG binding determines the fraction of active LacI

Before nondimensionalization, let x_{phys} denote the intracellular TMG concentration. Let R denote active LacI and R_T the total LacI concentration. A simplified cooperative binding model is



The formation rate of the TMG-bound complex is proportional to

$$k_{\text{on}} Rx_{\text{phys}}^n,$$

whereas dissociation occurs at rate

$$k_{\text{off}}[Rx_n].$$

Because this binding equilibrates rapidly, set the two rates equal:

$$k_{\text{on}} Rx_{\text{phys}}^n = k_{\text{off}}[Rx_n].$$

Define the TMG-LacI dissociation constant x_0 in concentration units through

$$x_0^n \equiv \frac{k_{\text{off}}}{k_{\text{on}}}.$$

Then

$$[Rx_n] = R \left(\frac{x_{\text{phys}}}{x_0} \right)^n.$$

Total LacI is conserved:

$$R_T = R + [Rx_n].$$

Therefore

$$\frac{R}{R_T} = \frac{1}{1 + (x_{\text{phys}}/x_0)^n}.$$

From this point onward, measure intracellular TMG in units of x_0 and reuse the symbol

$$x \equiv \frac{x_{\text{phys}}}{x_0}.$$

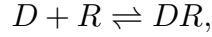
Then

$$\boxed{\frac{R}{R_T} = \frac{1}{1 + x^n}.$$

This is Eq. (3). The form is a simplified Hill description: intermediate states with fewer than n bound inducer molecules are not represented explicitly. A more mechanistic treatment can be obtained from an MWC model.

A.2 Active LacI occupancy determines promoter activity

Now consider binding of active LacI to the lac operator/promoter:



where D is free promoter and DR is repressor-bound promoter. At rapid binding equilibrium,

$$k'_{\text{on}}[D]R = k'_{\text{off}}[DR].$$

Define the LacI-operator dissociation constant

$$R_0 \equiv \frac{k'_{\text{off}}}{k'_{\text{on}}}.$$

Then

$$[DR] = [D] \frac{R}{R_0}.$$

The total promoter concentration is

$$D_T = [D] + [DR] = [D] \left(1 + \frac{R}{R_0} \right).$$

Therefore the fraction of promoter that is free is

$$\frac{[D]}{D_T} = \frac{1}{1 + R/R_0}.$$

If α is the maximal rescaled LacY production level when the promoter is free, production is proportional to this free-promoter probability:

$$P(R) = \alpha \frac{[D]}{D_T} = \frac{\alpha}{1 + R/R_0}.$$

This is Eq. (4).

A.3 Combining the two binding layers

Using the dimensionless intracellular TMG variable x defined above,

$$R = \frac{R_T}{1 + x^n}.$$

Therefore

$$P(x) = \frac{\alpha}{1 + \frac{R_T/R_0}{1 + x^n}}.$$

Define

$$\rho \equiv 1 + \frac{R_T}{R_0}.$$

Then

$$P(x) = \alpha \frac{1 + x^n}{\rho + x^n}.$$

Thus the sigmoidal production function used in the dynamical model is built from two molecular occupancy relationships:

TMG occupancy $\rightarrow$ fraction of active LacI $\rightarrow$ free-promoter probability $\rightarrow$ LacY production.

Appendix B. Glossary of terms

These are the terms mentioned in class. You should now understand what each of them means.

- Cell identity
- Lac operon
- All-or-none
- Memory
- Positive feedback
- Rate plot
- Phase / state space
- Vector field
- Fixed point
- Linear stability
- Basin of attraction
- Bistability
- Separatrix
- Bifurcation
- Bifurcation diagram
- Bifurcation point / fold point
- Hysteresis

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